

Weekly Note: Camera Exposure Factorial and FairFace7 ROI Gate on Pansor-20

FitSkin / Pansor

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Abstract

This week's work has two parts. **(A) Camera settings.** On a controlled ISO/shutter factorial ($N = 99$; four photographed subjects), chart-free ΔE_{00} under frozen **D65-FairFace7-ROI** tracks absolute exposure ($\text{corr}(\Delta E_{00}, EV_{\text{proxy}}) \approx 0.77$). Good cells (ISO ≈ 95 –111, shutter $\sim 1/250$ – $1/120$ s) give mean $\Delta E_{00} \approx 3.00$ – 5.00 ; long shutter ($\sim 1/60$ s) or ISO ≥ 200 push cheek L^* to ~ 78 – 85 and ΔE_{00} to ~ 15 – 21 . Pansor-20 is already fully in-band; an exposure residual helps the factorial in LOO but *hurts* Pansor and must not enter the color path. **(B) Non-exposure ROI.** Residual Pansor error was dominated by L^* ($\text{corr } 0.85$), led by Utsav (Indian misrouted Latino $\rightarrow$ Asian L_{p70} , $\Delta E_{00} \sim 8$ – 9). A FairFace-free-of-demographics gate (Latino + cheek $a^* \geq 11.5 + P(\text{SE Asian}) \geq 0.18 \rightarrow$ Indian ROI) remaps only Utsav and lowers cohort mean ΔE_{00} from 3.63 to 3.31 (Indian stratum 4.12 $\rightarrow$ 2.98). Frozen demosaic / $\sqrt{A \odot B}$ / affine / CAT are unchanged.

Takeaways.

1. Lock app AE: ISO ≤ 100 , shutter $\sim 1/120$ – $1/250$ s; reject ISO ≥ 200 , shutter $\geq 1/60$ s, $L^* \gtrsim 75$.
2. Do *not* ship the camera-settings Lab residual on Pansor.
3. Ship the Latino $\rightarrow$ Indian ROI gate with FairFace7 (default on): mean $\Delta E_{00} = 3.31$.
4. Next residuals: Luca / Dylan (White L^* / a^*), not exposure.

1 Part A — Camera exposure factorial

1.1 Motivation

D65-FairFace7-ROI recovers cheek CIELAB under D65 from indoor flash/no-flash ProRAW without a ColorChecker [1]. Pansor-20 mean ΔE_{00} assumes benign exposure. We ask: (1) how sensitive is the frozen path to ISO/shutter? (2) is Pansor already safe? (3) should an exposure residual enter Lab?

1.2 Methods (exposure)

Frozen color path. pre-AWB demosaic $\rightarrow$ cheek-median flash match $\rightarrow \mathbf{R} = \sqrt{\mathbf{A} \odot \mathbf{B}} \rightarrow \text{tier3_affine}$ (3×4) $\rightarrow$ Bradford CAT 5500 K $\rightarrow$ D65 $\rightarrow$ cheek Lab. Primary tables use frozen (off-path) ΔE_{00} unless noted.

Camera Settings Dataset. Pansor-app ProRAW zips; planned shutter/ISO/WB from **Camera Settings Aug10.xlsx**; FitSkin Inside Lab GT. WB locked at 6496 K. $N = 99$ trials. Emily: Lab only, no photos.

Table 1: Camera Settings subjects.

Person	FitSkin (L^*, a^*, b^*)	Letters	N zips	Notes
Giana	(60.49, 12.49, 16.98)	A–H	25	—
Keaton	(59.43, 14.92, 13.62)	A–H	25	—
Parker	(55.81, 11.36, 22.12)	A–H	22	missing E1
Wooj	(58.51, 13.29, 20.66)	A–H (+I)	27	I omitted from bar chart
Emily	(61.71, 12.59, 17.91)	A–H	0	Lab only

Design. A–D: shutter sweep (ISO roughly fixed). E–H: ISO sweep (shutter roughly fixed). $EV_{\text{proxy}} = \log_2(\text{ISO} \cdot t)$. E–H are well matched across people; A–D share shutter steps but baseline ISO differs slightly (Giana/Keaton 95 vs. Parker 111 / Wooj 116).

1.3 Results (exposure)

Table 2: Mean frozen ΔE_{00} and cheek L^* by letter ($N = 99$), ordered A–H. Blue = preferred; peach = failure. Wooj-only I omitted.

Letter	Role	Shutter	ISO	Mean ΔE_{00}	Mean L^*
A	Shutter (mid)	$\sim 1/130$ s	~ 95 –116	4.63	62.60
B	Shutter (long)	$\sim 1/60$ s	~ 95 –116	18.63	81.80
C	Shutter (faster)	$\sim 1/250$ s	~ 95 –116	5.42	53.40
D	Shutter (fastest)	$\sim 1/500$ s	~ 95 –116	7.31	51.20
E	ISO baseline	$\sim 1/130$ s	95	3.40	58.20
F	ISO 200	$\sim 1/130$ s	202	15.65	77.70
G	ISO ~ 400	$\sim 1/130$ s	~ 400	21.17	85.30
H	ISO ~ 1600	$\sim 1/130$ s	~ 1600	20.19	83.90

Overall factorial mean $\Delta E_{00} = 12.03$ (median 13.18). $\text{corr}(\Delta E_{00}, EV_{\text{proxy}}) \approx 0.77$, $\text{corr}(L^*, EV_{\text{proxy}}) \approx 0.87$.

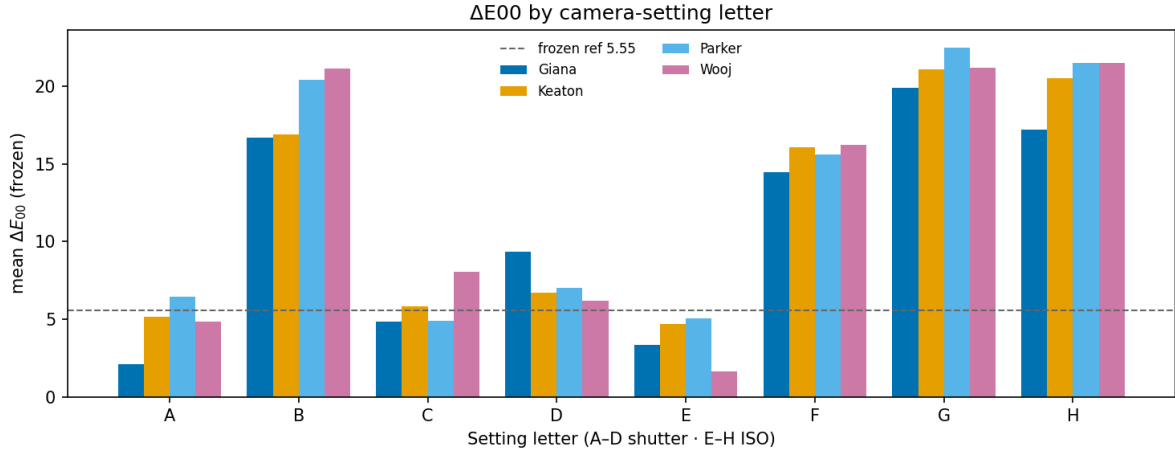


Figure 1: Mean ΔE_{00} by setting letter (A–H). Gray dashed: frozen trimmed-mean reference $\Delta E_{00} = 5.55$.

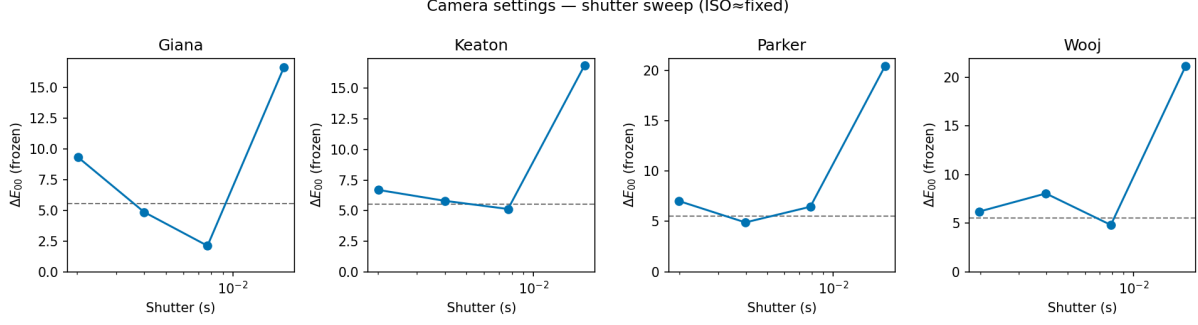


Figure 2: Shutter sweep (points sorted by ExposureTime). Cell B ($\sim 1/60$ s) spikes for every subject.

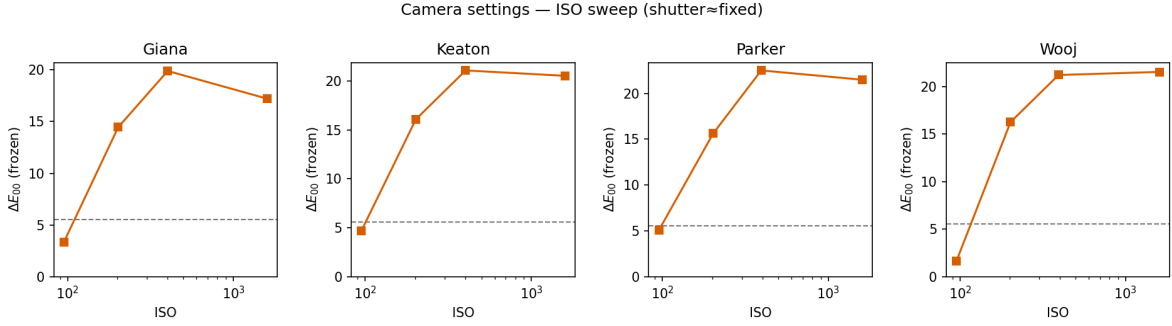


Figure 3: ISO sweep (points sorted by ISO). $\text{ISO} \geq 200$ (F–H) drives ΔE_{00} into the mid-teens–twenties.

Pansor exposure audit. All $N = 65$ FairFace7 indoor trials are in-band (ISO 50–125, shutter always 1/120s, L^* max ~ 66.6). A reject gate does not change today’s headline; it protects future collections.

Table 3: Pansor-20 no-flash EXIF audit ($N = 65$).

Quantity	Value
ISO (min / median / max)	50 / 64 / 125
Shutter	always 1/120 s
Out-of-band flags	0
In-band fraction	1.00

Exposure residual (rejected for Pansor). Camera-settings LOO: all-cells Lab residual 12.09→3.19; in-band L^* residual 5.17→2.87. Applied to Pansor: 3.63→7.40 (Black 3.34→17.97). **No Pansor win.** Keep residual out of the color path.

App capture policy. $\text{ISO} \leq 100$, shutter $\in [1/250, 1/120]$ s; reject $\text{ISO} \geq 200$, shutter $\geq 1/60$ s, or cheek $L^* \gtrsim 75$. Do not apply the four-person factorial Lab residual.

2 Part B — Non-exposure Pansor residuals (Latino→Indian gate)

2.1 Diagnosis

With exposure already in-band, residual ΔE_{00} correlates strongly with $|\Delta L^*|$ ($r \approx 0.85$). Worst subject: **Utsav** (Indian), FairFace=Latino_Hispanic→Asian low_C_Lp (L_{p70}), pipeline $L^* \approx 58$ –59

vs. FitSkin 49.65 ($\Delta E_{00} \sim 8-9$). Oracle demographics Indian `high_a_Lp` (L_{p35}) recovers $\Delta E_{00} \approx 1.91$. Blanket Latino→Indian hurts Eric (true Asian).

2.2 Gate (ROI only; default on)

If FairFace label = `Latino_Hispanic` **and** cheek $a^* \geq 11.5$ **and** $P(\text{Southeast Asian}) \geq 0.18$, route to **Indian** specular_tone rules. Remaps **only Utsav** (3 trials); Eric/Sanaz/others unchanged.

2.3 Results

Table 4: Pansor-20 FairFace7 mean ΔE_{00} before/after Latino→Indian gate ($N = 65$).

Stratum	n	Before	After
Overall	65	3.63	3.31
Indian	18	4.12	2.98
Asian / Black / Iranian / White	—	unchanged	unchanged
Utsav (3 trials)	3	~ 8.75	1.91

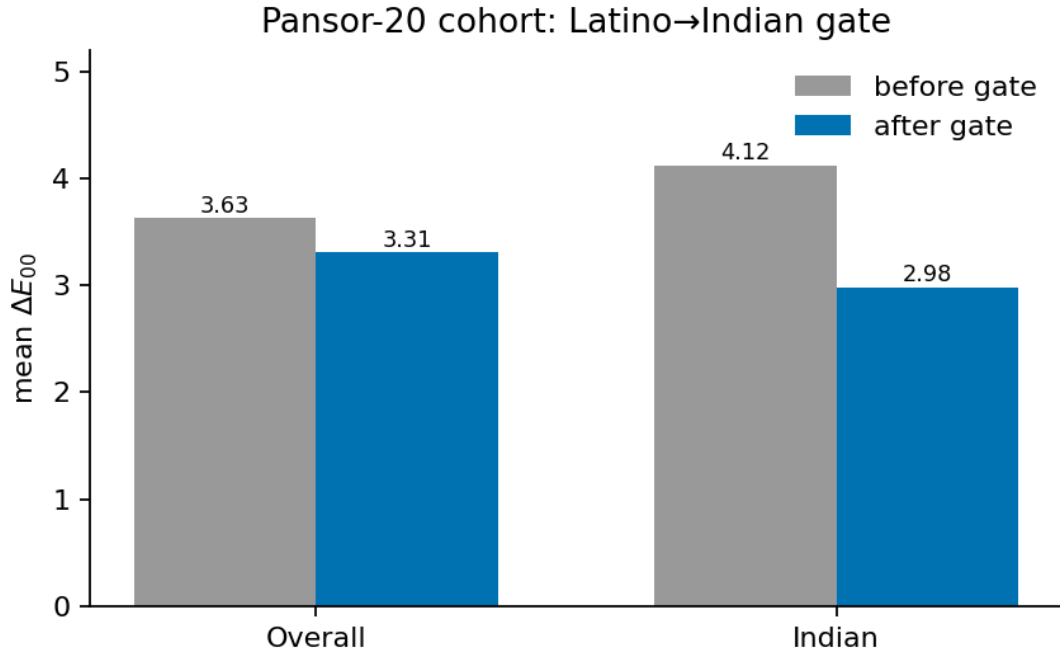


Figure 4: Cohort and Indian-stratum mean ΔE_{00} before vs. after the Latino→Indian gate.

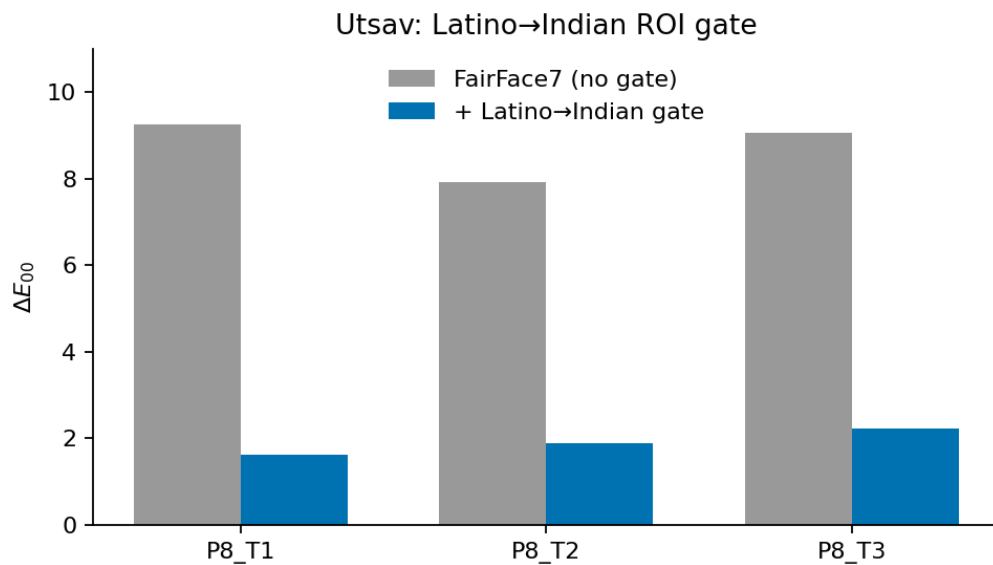


Figure 5: Utsav per-trial ΔE_{00} : FairFace7 without gate vs. with Latino→Indian gate.

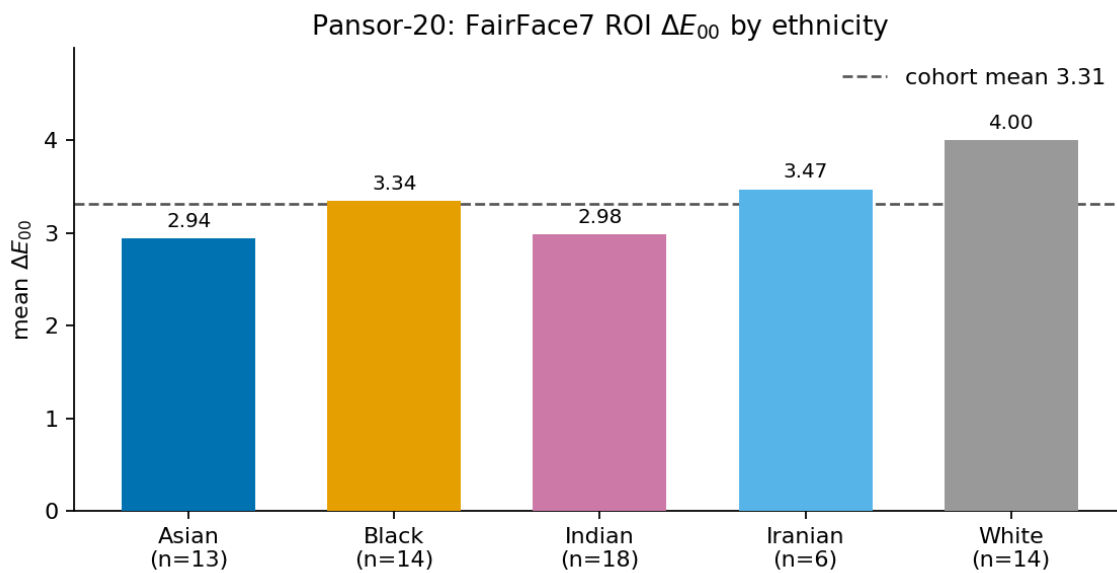


Figure 6: Mean ΔE_{00} by ethnicity after the gate (cohort mean 3.31).

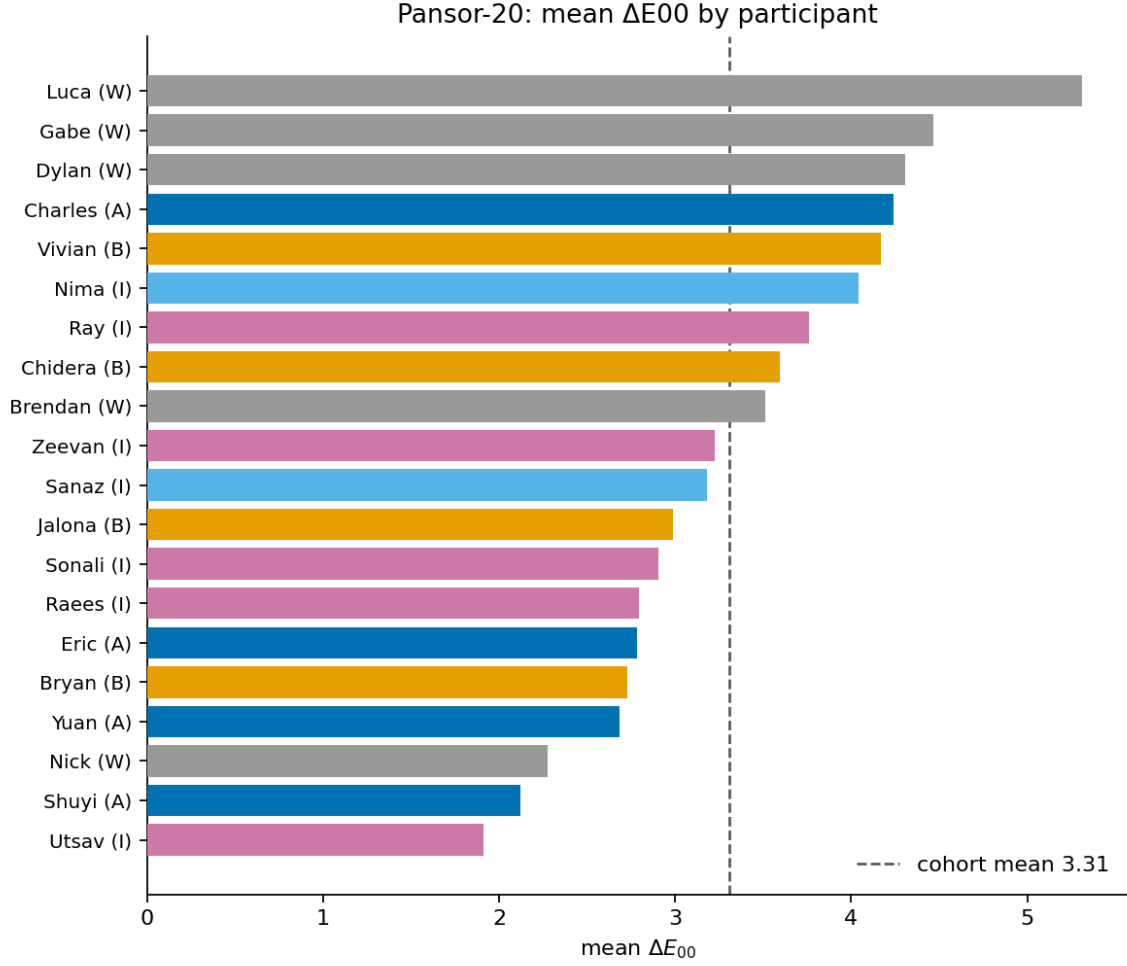


Figure 7: Mean ΔE_{00} by participant after the gate. Remaining highs: Luca, Dylan, Gabe (White).

Table 5: Mean / median ΔE_{00} by ethnicity after the gate.

Ethnicity	n	Mean ΔE_{00}	Median ΔE_{00}
Asian	13	2.94	2.65
Black	14	3.34	3.45
Indian	18	2.98	2.82
Iranian	6	3.47	3.63
White	14	4.00	3.84

2.4 Remaining non-exposure targets

- **Luca** (White): still too dark after L_{p70} ($\Delta E_{00} \sim 6.8$).
- **Dylan** (White): a^* too high / specular ($\Delta E_{00} \sim 5.7$); L-branch off ($L^* > 58$).
- **Ray / Gabe**: smaller residuals (~ 5).

3 Conclusion

Exposure control belongs in the **app** (AE lock + reject), not as a Lab residual on Pansor. On the in-band Pansor cohort, the largest residual was an ROI misroute (Utsav), fixed by a FairFace probability + cheek a^* gate that does not touch the frozen color path. Updated headline: mean $\Delta E_{00} = 3.31$ (median 3.18) with FairFace7 + Latino→Indian gate.

Reproduce.

```
# Camera settings
python scripts/evaluate_camera_settings.py --out-dir results/camera_settings

# Pansor FairFace7 + gate (default on)
python scripts/evaluate_pansor20_chartfree_d65.py \
  --scr-mode preawb_cat --fixed-cat-k 5500 --l-sampling fairface7 \
  --out-dir results/pansor20_fairface7
```

References

- [1] FitSkin technical note. *D65-FairFace7-ROI: Chart-Free Flash/No-Flash Skin Colorimetry with FairFace-Routed Cheek Sampling*. August 2026. (docs/d65_fairface7_roi_overleaf.tex).